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ETE 3351 Lab 4
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Conceptual analysis :

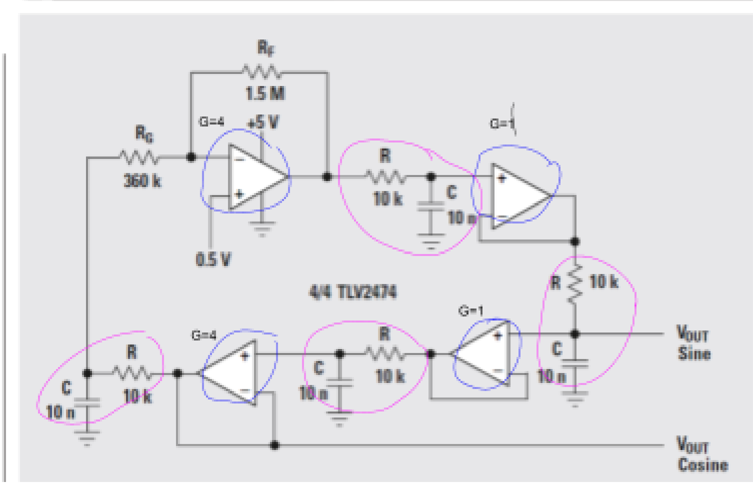
What is the bubba oscillator? The bubba oscillator is a type of circuit that takes in DC power and creates an AC signal. It is one of the easiest and most reliable oscillators to build and make operational. It converts DC power into an AC waveform. Every resistor and amplifier naturally produces tiny voltage fluctyatuins and at power up, these noise voltages kick the circuit with a very small signal. In most circuits, the noise dies away unless the barkhausen criteria is satisfied. To make an oscillator work, it needs to meet the Barkhausen criteria and needs two things

the loop must provide a total phase shift of 360. In thise case -180 from each RC stage, and another -180 from the op amps

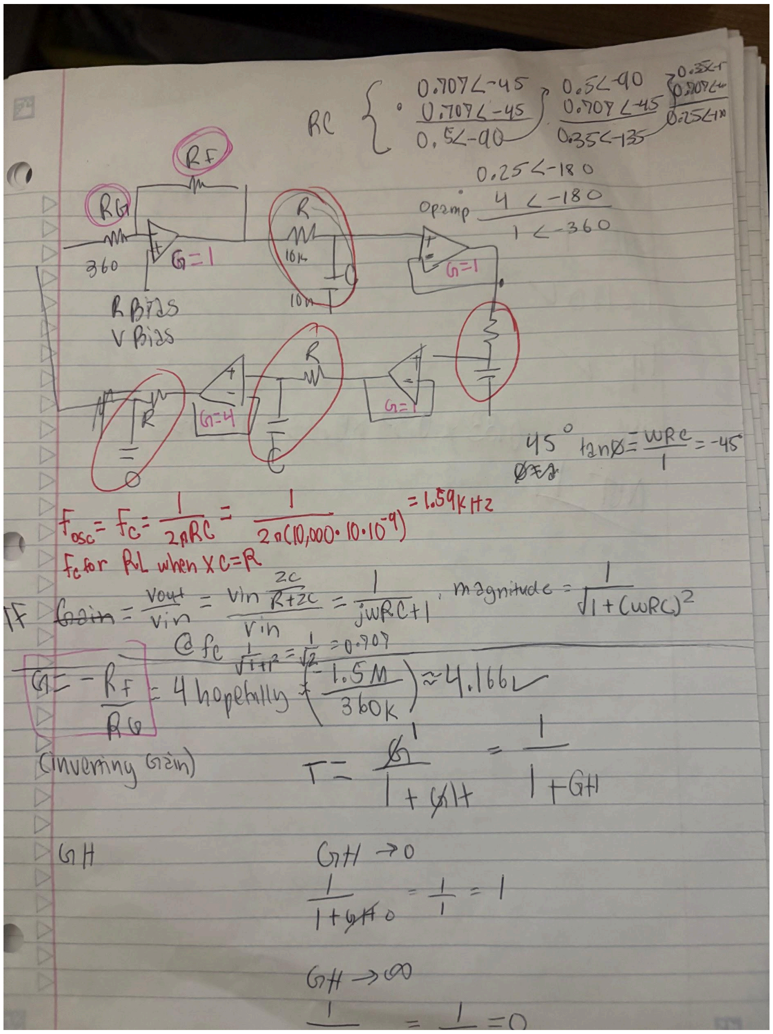
The loop must have an overal gain of one, so that the signal keeps going without fading or blowing up. In this case, the RC stages reduces amplitude to 25% of the input, and the op amp restores it by a factor of 4, resulting in a total loop gain of 1 so the signal survives and stabilizes.

Therefore, instead of vanishing, the noise is reinforced and shaped into a smooth sine wave.

Figure 9. Bubba oscillator



45 + 45 + 45 + 45 =180 0.707 * 0.707 * 0.707 * 0.707 = 0.25 = $\frac{1}{4}$



IMG 1. *Calculations of our circuit from lab*

Calculating oscillation frequency:
fosc = 1 / (2 * pi * R * C * sqrt(8))

Using the component values:
R = 10,000 ohms
C = 10 nF (10 x 10^-9 F)

fosc = 1 / (2 * pi * 10,000 * 10 x 10^-9 * sqrt(8))
fosc ≈ 1.59 kHz

Calculating the op amp gain
A = Rf / Rg
Therefore:
-Rf / Rg = -1.5MΩ/360kΩ= -4.166= 4

4<-180

Calculating the phase shift

The bubba oscillator uses four low pass RC networks, with a phase shift of 45 degrees each. We need 180 degrees, according to the Barkhausen criterion. We have: 4 LP filters, and 180/4 =45. In a transfer function, phases add: 45+45+45+45 = 180. The criterion is met.

In the lab and the problem in image 1, each RC stage has -45 degrees of phase shift

	0.707< -45	0.5<-90	0.35<-135	0.25<-180
*	0.707 < -45	* 0.707<-45	*0.707<-45	(op amp) *4<-180
	0.5<-90	0.35 < -135	0.25<-180	1<-360,

1<-360 satisfies Barkhausen phase condition

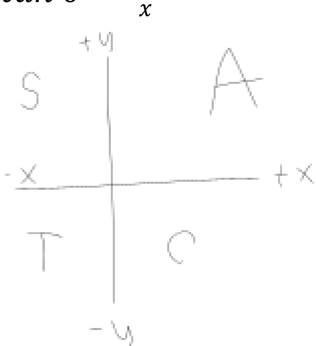
The 45 degree point is the fc or critical frequency of an RC lowpass filter network.

The signal will be reduced by ¼ because 0.707 * 0.707 * 0.707 * 0.707) = 0.25 = ¼.The four RC networks reduce the signal to 0.25 of its original strength. To cancel this loss, the amplifier multiplies by about 4 (1/0.25 =4).Therefore, the loop gain must be 4 * 0.25 = 1 to make the sine wave sustain and not shrink or blow up. In the lab, the inverted gain (calculated as -Rf/Rg) is -1.5MΩ/360kΩ or -4.166, which is sufficient. Gain is how much an amplifier multiplies a signal. Since each RC filter makes the signal smaller we need to use an amplifier to compensate . In this case we would use 4 op amps- one has a gain of 4 (which will be used last) and the others have a gain of 1.

How do we get 45 degrees at F cuttoff?

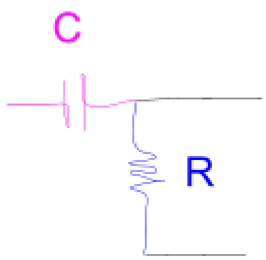
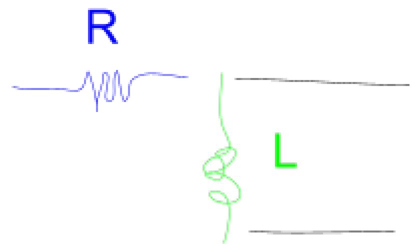
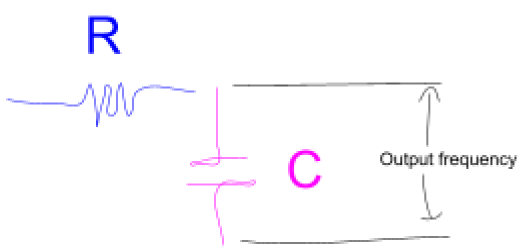
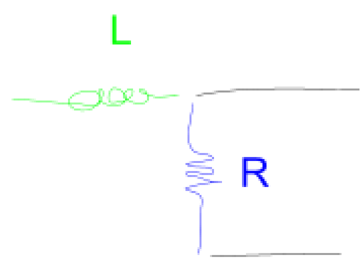
Recall that a RC LPF has a phase of -45 degrees. As far as calculations go,

$tan\,\theta = \frac{y}{x}$



$$\theta = arctan\frac{imaginary\,part)}{(real\,part)} \;=\; -\,arctan\,\frac{\omega RC}{1} \;=-\,45$$

	RC	RL

	$*Fc = Fo = \frac{1}{2\pi RC}$	$Fc = Fo = \frac{R}{2\pi L}$
High pass (passes high frequency, blocks low frequency)		
Low Pass (passes low frequency, blocks high frequency)		

*Calculations:

Fc for XL

$XL = R$, $\omega_C L = R$, $2\pi f_C L = R$, $f_c = \frac{R}{2\pi L}$

Fc for XC

$XC = R$, $\frac{1}{\omega_C C} = R$, $\frac{1}{2\pi f_c C} = R$, $f_c = \frac{1}{R\pi 2C}$

The role of the potentiometer is to allow fine tuning of gain to keep the oscillations stable. The oscillator doesnt need external input- it is like a swing with no air resistance. The oscillator pushes itself using feedback so it doesnt need an external push (input) and is self sustaining. However, it does need a start up voltage.

The key closed loop transfer function equation for a negative feedback system such as the bubba oscillator:

$TF = \frac{1}{1+GH}$

This formula shows how controlling the feedback (GH) affects the overall transfer function in a closed negative feedback loop.

TF= transfer function (how the output of the system relates to its input. The TF tells me " if I apply some input, what fraction or multiple shows up at the output after feedback?")

G = gain of the system
H= the feedback factor (how much of the output is fed back to compare with the input)
GH= loop gain (total gain around the loop)
*assume that G= 1 for both cases

CASE 1: GH = approximately 0
Since G*H = 0, and gain =1, H= approximately 0, meaning that the feedback factor is approximately 0

$TF = \frac{1}{1+GH} = \frac{1}{1+0} = 1$

This means that Vout is approx Vin, the output is approx equal to the input, so feedback has almost no effect.

CASE 2 : GH = approximately infinity
SInce G*H= infinity and G=1, feedback factor (H) is approx infinity. This means that

$TF = \frac{1}{1+GH} = \frac{1}{1+\infty} = 0$

IN this case, the closed loop system output approaches zero, because the strong negative feedback suppresses the signal, and the waveform therefore dies out instead of sustaining.

*Negative feedback with large GH makes the output wave form fade to 0
*Negative feedback with small GH means small H and therefore TF=1, and results in a sustained oscillation and meets Barkhausen criterion.

What does changing R and C do? Since $f_{osc} = \frac{1}{R\pi 2C}$, R and C are inversely proportional to R & C. Increasing R and/or C increases denominator and makes the frequency smaller, decreasing R and/or C decreases denominator and makes frequency bigger. Recall that f_{osc} is the rate at which the output waveform repeats itself in an oscillator, measured in cycles per second (Hz), in a bubba oscillator this is the frequency of the sine wave that keeps coming out once the circuit is running steadily. So if the bubba oscillator frequency is 1.6Hz, that means that the sine wave completes 1,600 full cycles every second and each cycle lasts 1/1600 or 625 micro seconds. Lowering frequency means more time to complete a cycle, wave is slower, slower oscillation, so therefore R and C control oscillation speed by controlling the frequency.

Where does 0.707 come from? The value 0.707 comes from the definition of the cutoff frequency (-3dB frequency) of an RC filter.

$GAIN = \frac{V\,out}{V\,in}$

The resistor is in series with the capacitor, and the output is taken across the capacitor only. Therefore we can use a voltage divider to find the voltage across the capacitor

$Vout = Vin\,(\frac{ZC}{R+ZC})$

So

$GAIN = \frac{V\,out}{V\,in} = \frac{Vin\,(\frac{ZC}{R+ZC})}{Vin} = \frac{Zc}{R+Zc} = \frac{\frac{1}{wC}(-j)}{R+\frac{1}{wc}(-j)}$

Simplify

$\frac{\frac{1}{wCj}}{R+\frac{1}{wc}j} * \frac{j\omega C}{j\omega C} = \frac{1}{j\omega RC+1}$

This is the transfer function: $\frac{1}{j\omega RC+1}$

However, we want the *magnitude* of the transfer function : $\sqrt{(\frac{1}{j\omega RC+1})^2} = \frac{1}{\sqrt{(j\omega RC+1)^2}}$

$magnitude\ of\ GAIN = \frac{V\,out}{V\,in} = \frac{1}{\sqrt{1+(wRC)^2}}$

The cuttoff frequency is defined when $\omega_c = \frac{1}{RC}$

(Fc for XC = $XC = R$, $\frac{1}{\omega_c C} = R$, $\frac{1}{2\pi f_c C} = R$, $f_c = \frac{1}{R\pi 2C}$)

Therefore,

$magnitude\ of\ GAIN\ at\ F_c = \frac{V\,out}{V\,in} = \frac{1}{\sqrt{1+(1)^2}} = \frac{1}{\sqrt{2}} = 0.707$

*$magnitude\ of\ GAIN = \frac{V\,out}{V\,in} = \frac{1}{\sqrt{1+(wRC)^2}}$

At Fc ($\omega_c = \frac{1}{RC}$)---> $\frac{V\,out}{V\,in} = \frac{1}{\sqrt{1+(wRC)^2}} = \frac{1}{\sqrt{1+(1)^2}} = \frac{1}{\sqrt{2}} = 0.707$

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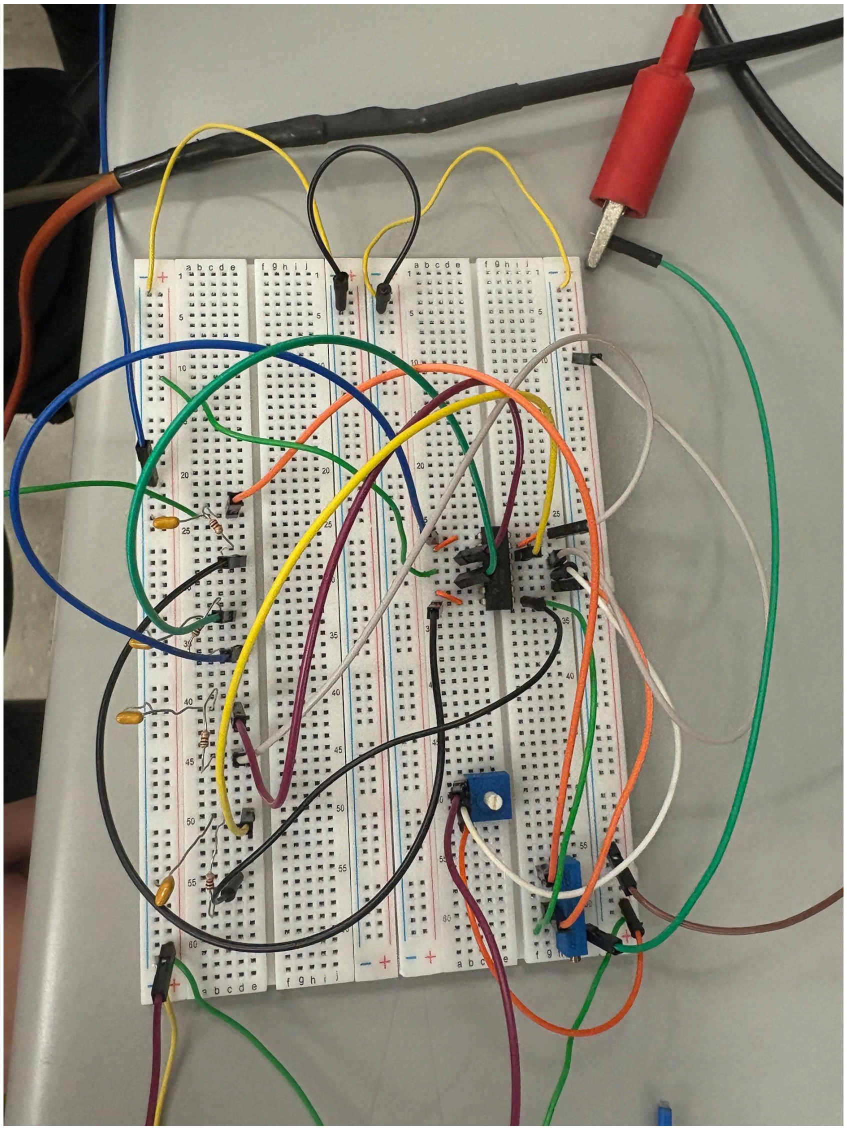
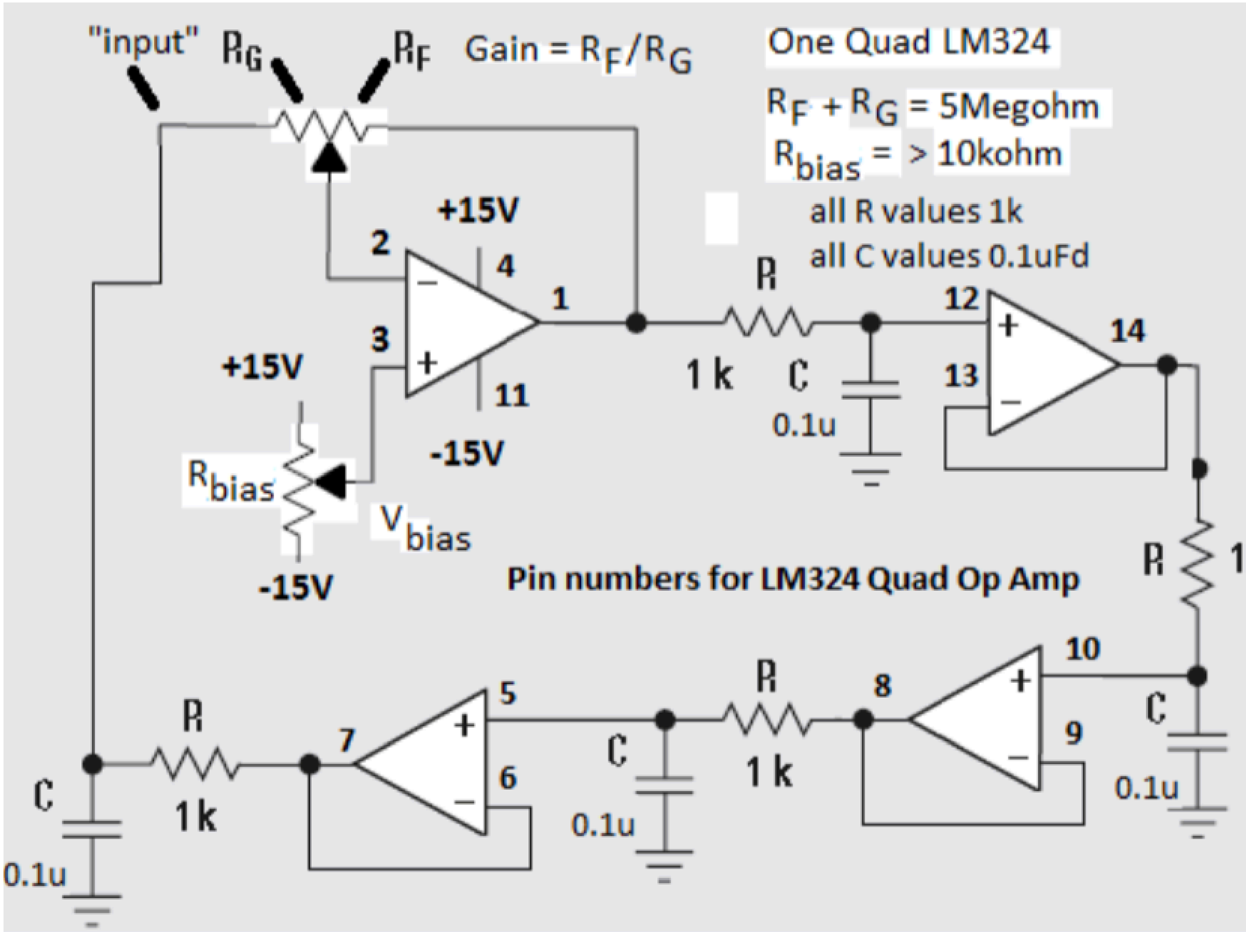
Part 1:

1. Construct the oscillator circuit, see figure below for suggested design. You might find it easier to use 4 single op amps like the LM741, both it and quad op amp designs work, but the pin numbers will change for the four single op amps.



IMG 2. *The circuit*



IMG 3. *Schematic*

2. Preset the Gain pot, i.e. 5Meg unit, to 4 = RF/RG = 4Meg/1Meg, it is also useful to know which way turning the pot (CW or CCW) increases the gain

The RC phase-shift oscillator circuit we built uses an op-amp with a required gain of approximately 4 and produces an oscillation frequency close to 1.6 kHz. This gain requirement comes from the RC network attenuation that was shown in previous calculations, and we found that each RC stage attenuates by about 0.707, so multiplying across the three stages gives approximately 0.25 of the input signal. To compensate the amplifier must provide a gain of 4 (1/0.25). Each RC stage also contributes roughly 45° of phase shift at the oscillation frequency—three stages produces 135°, and the inverting amplifier adds the remaining 180°, reaching the Barkhausen phase requirement of 360° total. If the gain is too high, the circuit becomes unstable and the sine wave distorts into a square wave; if the gain is too low, oscillations die out.

During testing, we measured the transfer function of each feedback block and adjusted the RG/RF potentiometer to set the correct oscillation behavior. Very high resistance caused the circuit leads to act like antennas, occasionally picking up radio noise. Additionally, the op-amp output should never be tied directly to ground, since this can damage the device. We also used Vbias of about 1 V to keep the output waveform centered properly. One practical issue encountered in the lab was that the potentiometers were very sensitive, making fine adjustment difficult. Overall, the experiment confirmed that correct gain, phase shift, and component values are needed to achieve stable oscillation at the expected frequency. Our kickstart voltage was 10.81v.

3. When you first power up the circuit you will monitor the output signal of the Gain op-amp (pin 1 in diagram) with your scope and Vbias, pin 3 of negative Gain op amp with DC voltmeter.

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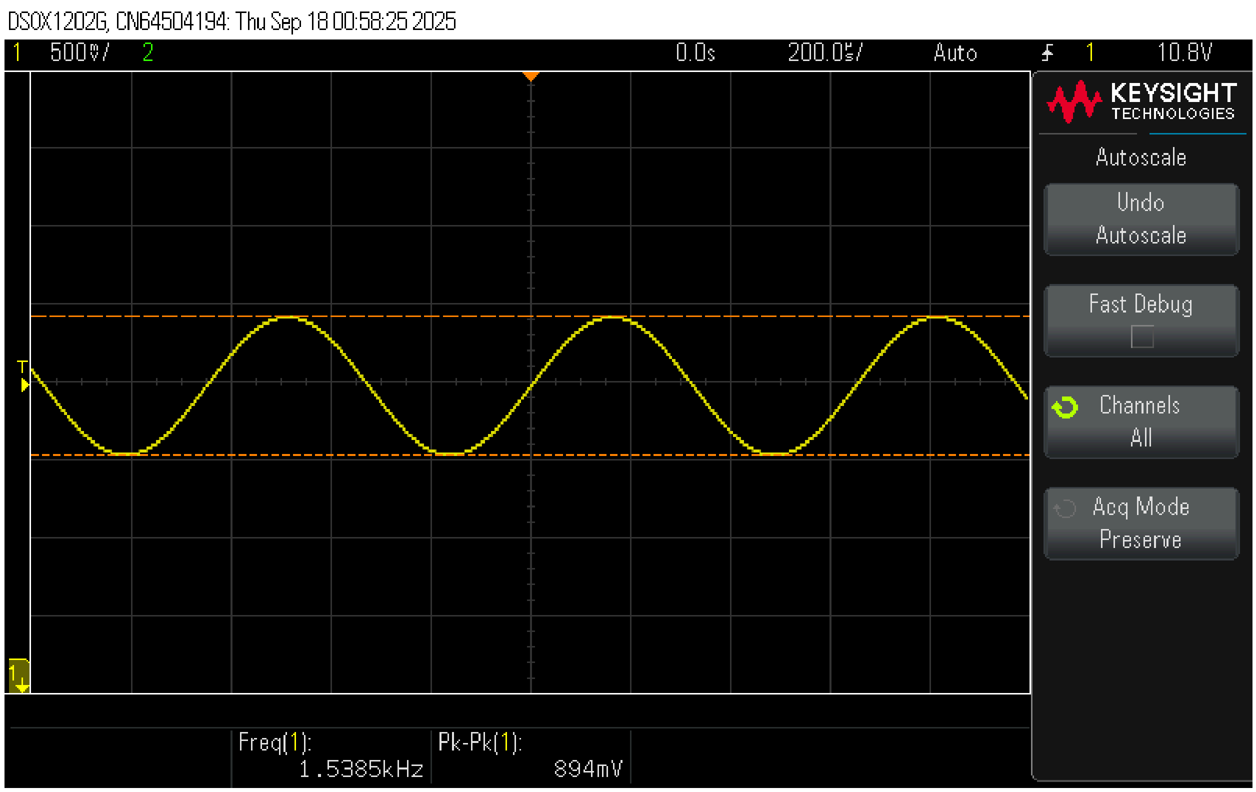
4. Power on both + and - supplies on and check power supply current supplies which

should be less than 10mA if using +- 15V supplies, turn off power supply and recheck the circuit wiring if higher currents >20mA are seen, which likely indicates a wiring error.

5. If no oscillation is present on Pin 1 of the op amp output adjust the gain upwards (RF increase, RG decrease) while alternatively adjust Vbias, repeat gain and Vbias adjustments until sustained oscillation is achieved.

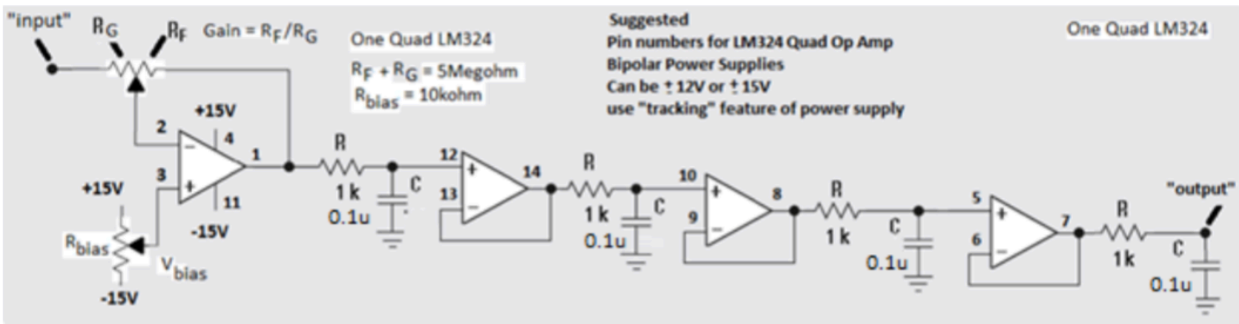
6. Check the frequency fOSC which should be close to 1.6kHz +- 100 Hz. Record the oscillator frequency, fOSC = 1.538khz
This matches the calculations shown in image 1 and explained in my conceptual analysis.



IMG 4.

7. Additionally for your oscillator record
a) oscillator voltage VOSC_pp = 2.8vpp,
b) Vbias = 1 V,
c) RF and RG values, remove the 5 Meg gain pot to measure each portion. -1.5MΩ /360kΩ or -4.166
RF = -1.5M ohms and RG = 360k ohms
The calculated gain= RF/RG = 1.5MΩ/360kΩ= 4.166

Part 2:
Verification of Barkhausen Criterion, Phase Measurement Section: To make measurements easier we will “break” the feedback loop and use an ARB test signal as ann“input” test signal as show



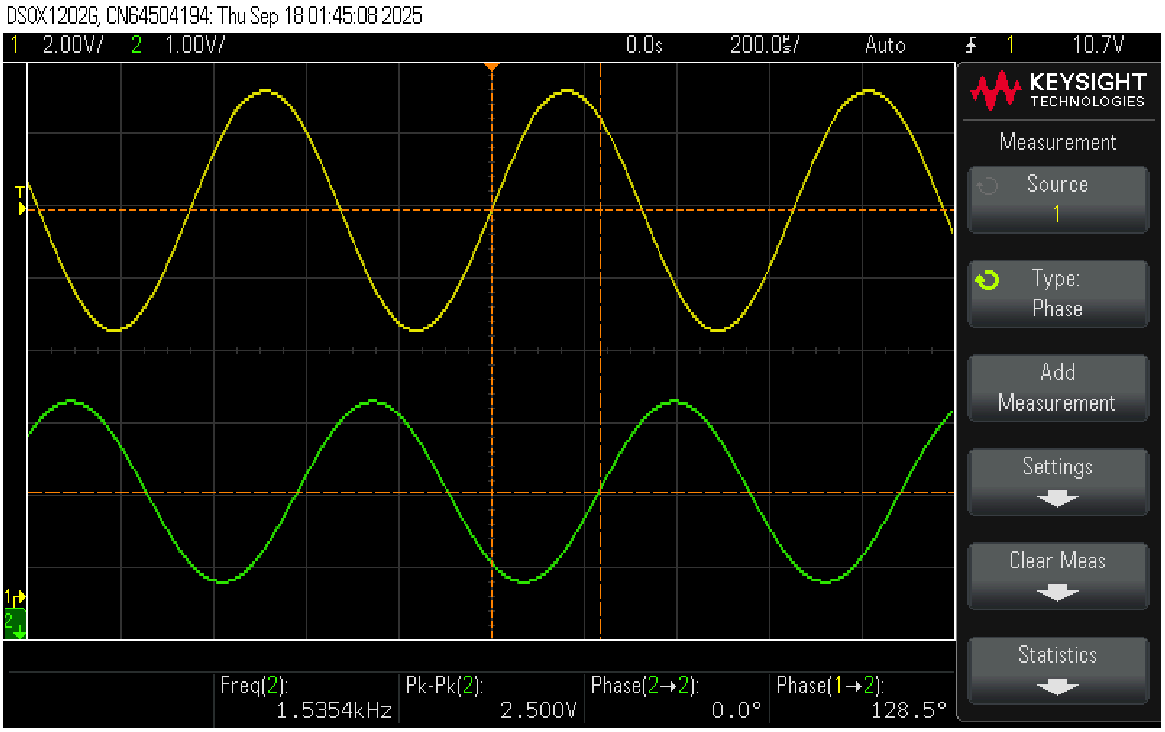
IMG 5.

9. Setup an ARB waveform generator as an “input” signal to this cascaded amplifier string. The ARB signal should be a sine wave, adjust its frequency and magnitude to match yournsine wave Oscillator values from steps 6 and 7, frequency fOSC and VPP.

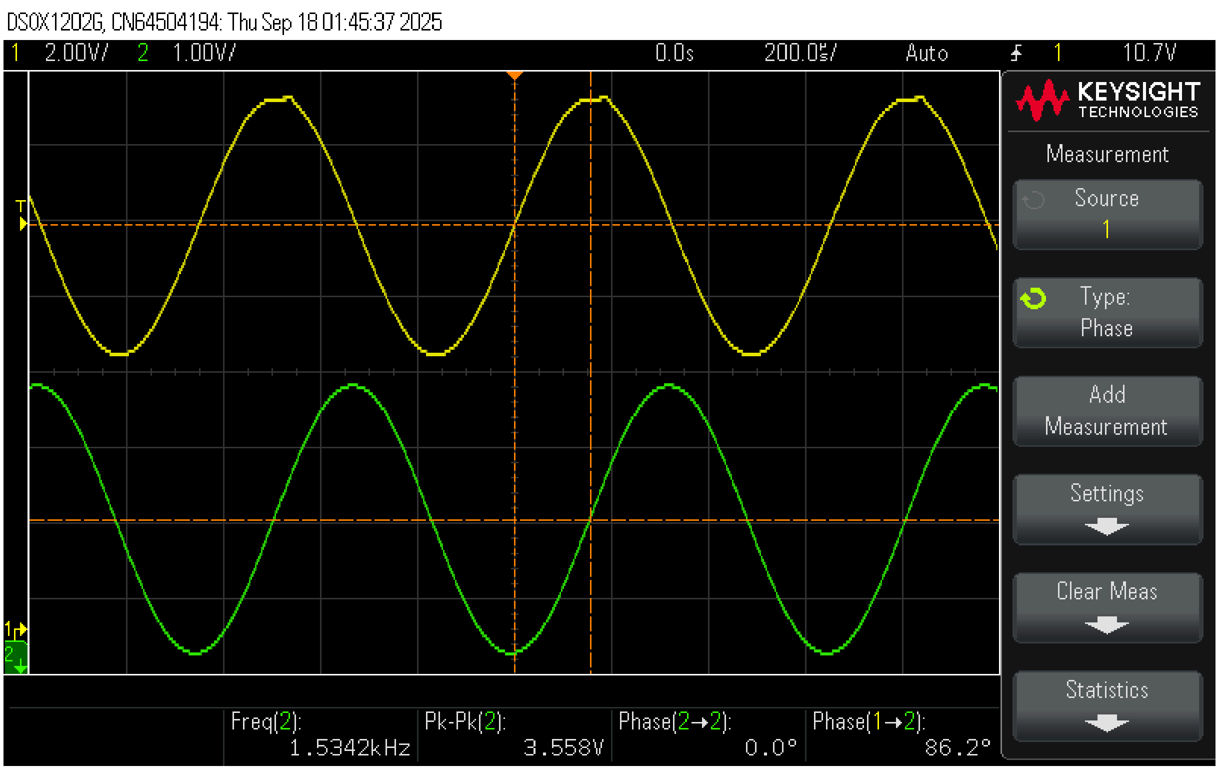
For this step Channel 2 of the oscilloscope was connected to measure the output voltage of the oscillator while sharing a common ground reference with the circuit. Channel 1 was used to measure the input side of the potentiometer (at pin 1 of the op-amp) so that the transfer function could be observed. By temporarily breaking the feedback loop, we were able to measure how the circuit responds to the input signal rather than allowing the oscillator to freely run. This setup allowed us to verify that the phase shift between the input and output was correct and that the gain and feedback conditions were properly adjusted to support oscillation.

10. View the output signal and if necessary reduce the op amp gain to minimize any distortion in your output signal sine wave, adjust ARB signal magnitude only if necessary.

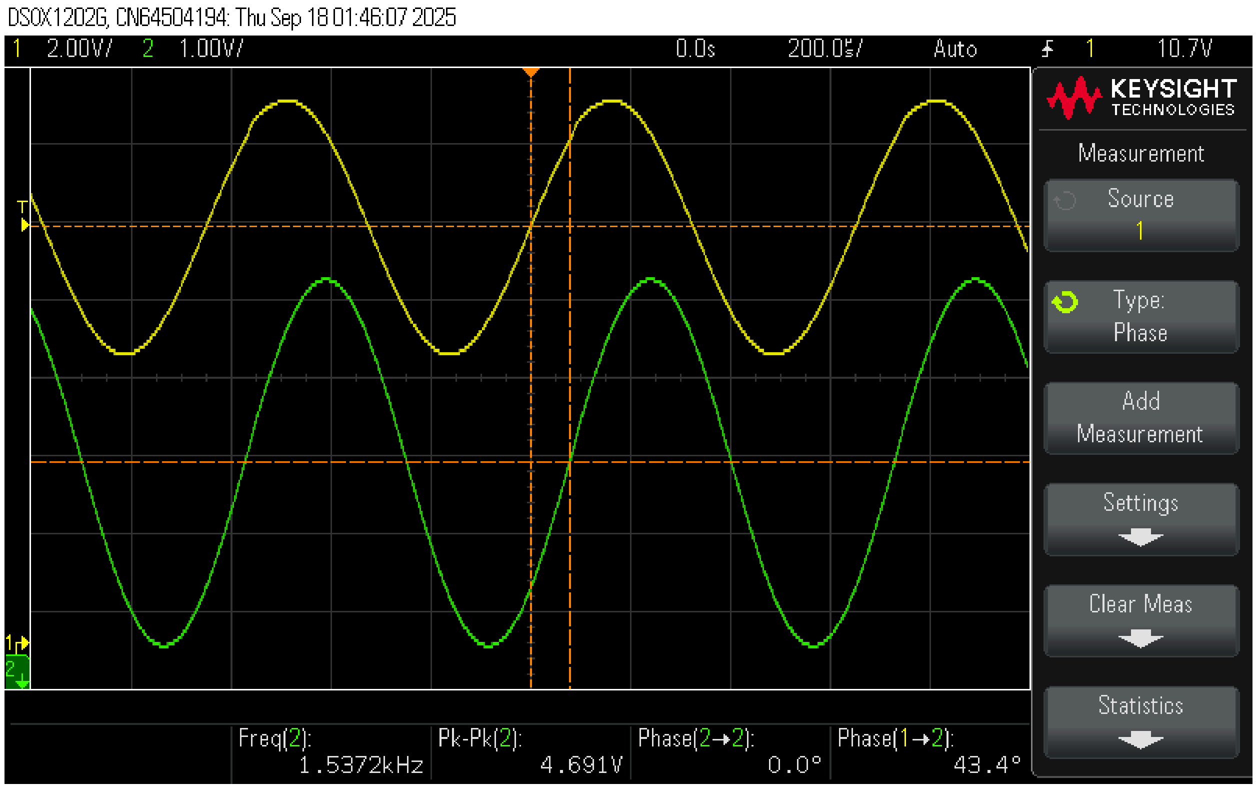
11. Record Transfer function values (magnitude and phase) of each op amp output (pins 1, 14, 8 and 5) vs. input ARB signal. Confirm the Barkhausen criterion i.e. output = input, gain/phase = 1/0o


DSOX1202G, CN64504194: Thu Sep 18 01:45:08 2025
1 2.00V/ 2 1.00V/ 0.0s 200.0us/ Auto 1 10.7V
KEYSIGHT TECHNOLOGIES
Measurement
Source 1
Type: Phase
Add Measurement
Settings
Clear Meas
Statistics
Freq(2): 1.5354kHz
Pk-Pk(2): 2.500V
Phase(2→2): 0.0°
Phase(1→2): 128.5°

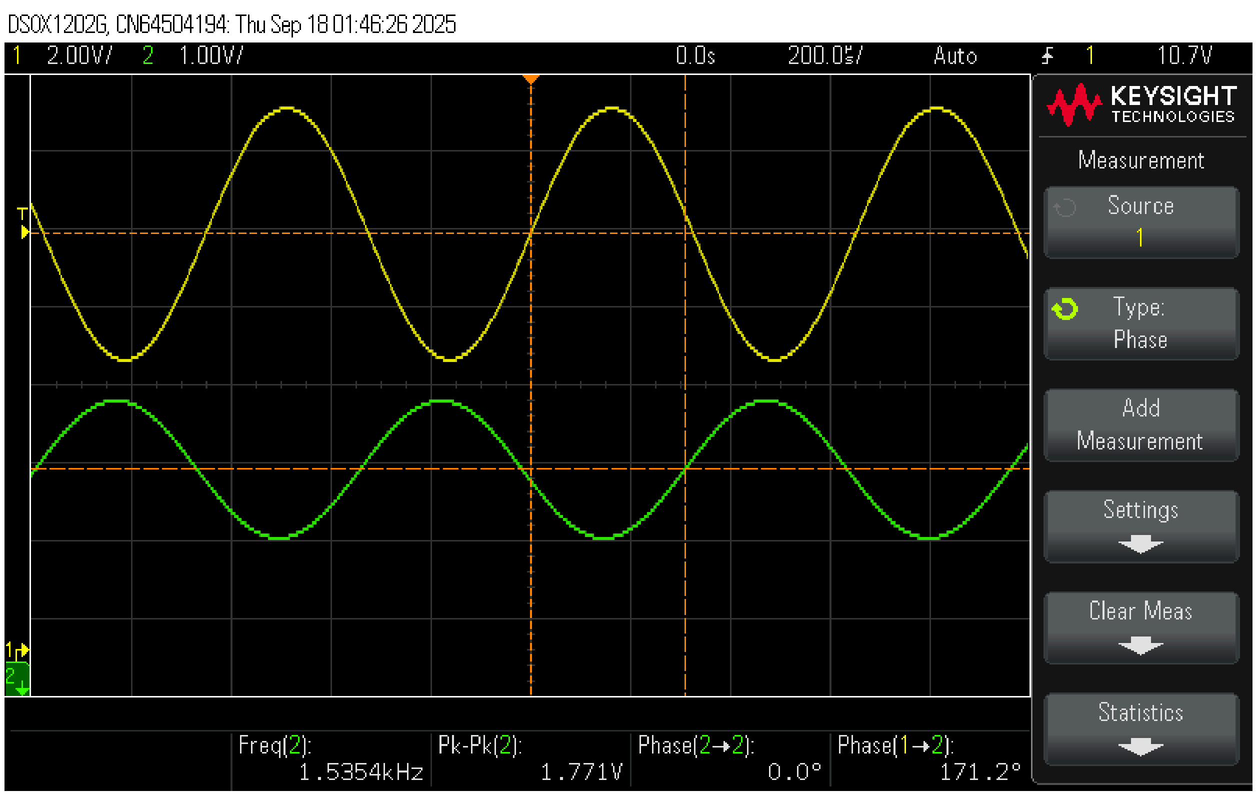
IMG 6.


DSOX1202G, CN64504194: Thu Sep 18 01:45:37 2025
1 2.00V/ 2 1.00V/ 0.0s 200.0us/ Auto 1 10.7V
KEYSIGHT TECHNOLOGIES
Measurement
Source 1
Type: Phase
Add Measurement
Settings
Clear Meas
Statistics
Freq(2): 1.5342kHz
Pk-Pk(2): 3.558V
Phase(2→2): 0.0°
Phase(1→2): 86.2°

IMG 7.


DSOX1202G, CN64504194: Thu Sep 18 01:46:07 2025
1 2.00V/ 2 1.00V/
0.0s 200.0㎲/ Auto 1 10.7V
KEYSIGHT TECHNOLOGIES
Measurement
Source 1
Type: Phase
Add Measurement
Settings
Clear Meas
Statistics
Freq(2): 1.5372kHz
Pk-Pk(2): 4.691V
Phase(2→2): 0.0°
Phase(1→2): 43.4°

IMG 8.


DSOX1202G, CN64504194: Thu Sep 18 01:46:26 2025
1 2.00V/ 2 1.00V/
0.0s 200.0㎲/ Auto 1 10.7V
KEYSIGHT TECHNOLOGIES
Measurement
Source 1
Type: Phase
Add Measurement
Settings
Clear Meas
Statistics
Freq(2): 1.5354kHz
Pk-Pk(2): 1.771V
Phase(2→2): 0.0°
Phase(1→2): 171.2°

IMG 9.